

Hospital Website Deployment using AWS EC2 & Docker Compose

- This project demonstrates the deployment of a Hospital Website on an AWS EC2 instance using Docker Compose.
- The application is containerized with Docker and managed through Docker Compose, enabling easy deployment and scalability.

Features

- Responsive Hospital Website
- Dockerized Flask Application
- Multi-stage Docker Build
- Docker Compose Deployment
- AWS EC2 Hosting
- Security Group Configuration
- Docker Network & Volume Management
- Accessible via EC2 Public IP

Technologies Used

- AWS EC2
- Docker
- Docker Compose
- Flask (Python)
- Ubuntu
- Linux
- Git & GitHub

Deployment Workflow

1. Launch an AWS EC2 Ubuntu instance.
2. Install Docker and Docker Compose.
3. Clone the GitHub repository.
4. Build Docker images using Docker Compose.
5. Create Docker network and volumes.
6. Start the application using:

`docker compose up -d`

7. Configure the EC2 Security Group to allow HTTP traffic (Port 5000 or 80).
8. Access the Hospital Website through the EC2 Public IP.

Learning Outcomes

- Containerization using Docker
- Multi-container deployment with Docker Compose
- AWS EC2 instance management
- Docker networking and persistent volumes
- Cloud application deployment
- Linux command-line operations
- DevOps deployment workflow

```
https://us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressfamily=ipv4&connType=standard&iinstanceId=i-055f85a6cf6963ac0&osUser=ubuntu&region=us-east-1

[flask-app build 2/4] WORKDIR /app
[flask-app build 3/4] COPY app.py .
[flask-app build 4/4] RUN pip install --no-cache-dir flask
[flask-app stage-1 3/4] COPY --from=build /usr/local/lib/python3.9 /usr/local/lib/python3.9
[flask-app stage-1 4/4] COPY --from=build /app /app
[flask-app] exporting to image
=> exporting layers
=> exporting manifest sha256:c2a08a8bcbec309dd3f8384a220c547fc46c3f5d34e96941653cdc3d5452897e8
=> exporting config sha256:a7b2d7b7c3be55a393f6979afc256397ee77a5fe24ce71a3b386445ebcb524
=> exporting attestation manifest sha256:b6040d856fe233a27e15875bd7d204220d2c35647f1077c7e20e8b9bacf2e8be
=> exporting manifest list sha256:d3edcd8dd003b2e9e476f23ff7b79e93c0af380f255f9909cfc74658c1750f
=> naming to docker.io/library/ubuntu-flask-app:latest
=> pushing to docker.io/library/ubuntu-flask-app:latest
[flask-app] resolving provenance for metadata file
[+] Running 4/4
  flask-app      Built
  Network ubuntu_default      Created
  Volume ubuntu_flask-volume  Created
  Container flask-container    Created
Attaching to flask-container
flask-container | * Serving Flask app 'app'
flask-container | * Debug mode: off
flask-container | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
flask-container | * Running on all addresses (0.0.0.0)
flask-container | * Running on http://127.0.0.1:5000
flask-container | * Running on http://172.18.0.2:5000
flask-container | Press Ctrl-C to quit
flask-container | 103.183.240.10 - - [22/Jul/2026 06:48:58] "GET / HTTP/1.1" 200 -
flask-container | 103.183.240.10 - - [22/Jul/2026 06:48:59] "GET /favicon.ico HTTP/1.1" 404 -
[+] Enable Watch

i-055f85a6cf6963ac0 (indhu)
PublicIPs: 18.234.51.110 PrivateIPs: 172.31.23.127
CloudShell Agent Toolkit for AWS Feedback Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences
```

```
https://us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressfamily=ipv4&connType=standard&iinstanceId=i-055f85a6cf6963ac0&osUser=ubuntu&region=us-east-1

Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Jul 22 06:47:45 UTC 2026

System load: 0.0      Temperature:   -273.1 C
Usage of /:  38.9% of 6.61GB   Processes:    168
Memory usage: 9%      Users logged in: 0
Swap usage:  0%        IPv4 address for emp39s0: 172.31.23.127

Expanded Security Maintenance for Applications is not enabled.

77 updates can be applied immediately.
53 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Jul 22 06:39:02 2026 from 18.206.107.28
ubuntu@ip-172-31-23-127:~$ sudo su
root@ip-172-31-23-127:/home/ubuntu# ls
Dockerfile  app.py  compose.yml
root@ip-172-31-23-127:/home/ubuntu# docker compose up
WARN[0000] Docker Compose is configured to build using Bake, but buildx isn't installed
[+] Building 6.0s (12/12) FINISHED
=> [flask-app internal] load build definition from Dockerfile
=> transferring Dockerfile: 709B
=> [flask-app internal] load metadata for docker.io/library/python:3.9-slim
docker:default
0.08
0.08
0.42

i-055f85a6cf6963ac0 (indhu)
PublicIPs: 18.234.51.110 PrivateIPs: 172.31.23.127
```



